E31: the chain solve for the $k = 4$ V_4-line decomposition

Goal G2, `RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` §9.22 (pre-registration G2.85–G2.94), §9.23 (results) · 2026-10-24
Result class: engineering (phase speed-up) · binary `a8f52422d5`

1. Result

E31 adds a second solver for one $k = 4$ decomposition on the V_4 line (the degree-11 system in the invariants of $V_4^{4_0} \rtimes S_4$), beside E25's unchanged Macaulay solver. It works in the quotient as a 12-generator module over $\mathbb{F}_p[P_2]$: left-looking normal forms of the 12 chain ends, $\det R(x)$ of a 12×12 polynomial matrix, kernels of $R(\lambda)$. **A** is that solver; **B** is A with the sparsest pivot rows in the echelon. On the same 10,000 calls (Stage A, 10 curves) both returned the existing solver's decompositions on every call (10,000 / 10,000 and 10,000 / 10,000), and the per-call median C fell by **2.62–2.69× (A)** and **4.99–5.32× (B)** counted $\mathbb{F}_p$ multiplications.

End to end on E25's ten curves, both arms, the same target streams (40 records), every relation, logarithm and walk is unchanged and S falls by **2.50–2.56× (A)** and **4.53–4.76× (B)** at 0 GAE. Priced at the §5a band (one GAE = 97 $\mathbb{F}_p$ multiplications per Macaulay-matrix entry and per walk step), the factor is 1.39–1.43 (A) and 1.59–1.64 (B) at 0.1 GAE, and 1.034–1.053 (A) and 1.051–1.071 (B) at 1 GAE: E31 stays below E25 on every row at every weight (yes). What E31 cut is the arithmetic, not the Macaulay matrix the solve still builds; the exponents are unchanged. **Class: engineering (phase speed-up)**, a verified, measured improvement of the index-calculus baseline under the 2026-10-15 criterion.

Distance remaining. The matched $\langle \pm\psi \rangle$ rho is still ahead of every row of both solvers at every weight (yes). At 0 GAE, B's rows are $2.39 \cdot 10^3$–$1.52 \cdot 10^5\times$ the matched rho (best $2.39 \cdot 10^3\times$ at $p = 397$ seed 1, arm B; E25's best on the same curve was $1.13 \cdot 10^4\times$); at 0.1 GAE $1.31 \cdot 10^4$–$8.55 \cdot 10^5\times$; at 1 GAE $7.21 \cdot 10^4$–$5.14 \cdot 10^6\times$.

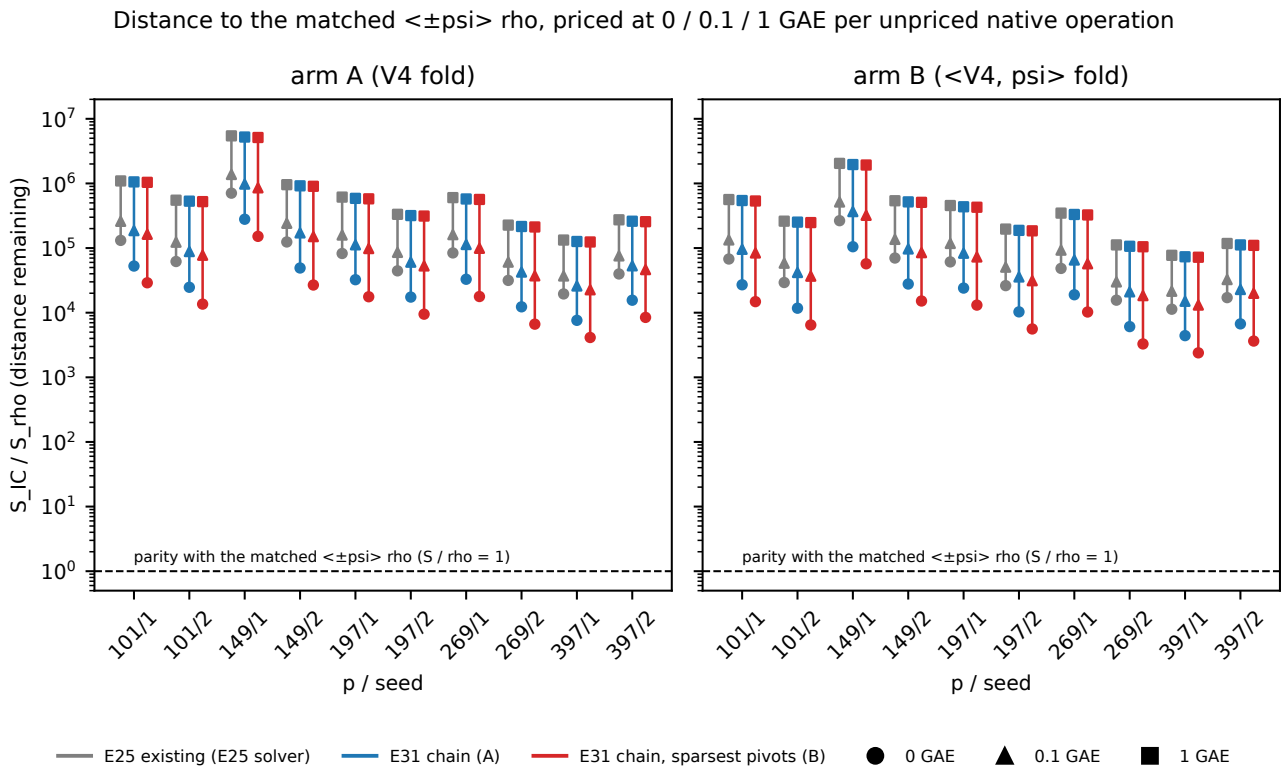


Figure 1: S_{IC}/S_ρ per curve and arm, E25 (existing solver) against E31 A and B, each at 0 (circle), 0.1 (triangle) and 1 GAE (square) per unpriced native operation. S_ρ is the mean of 16 matched $\langle \pm\psi \rangle$ walks per curve (`rho_matched_muls`, `rho_matched_steps`). Records: `experiments/30_gls_v4/p*s*.json`, `experiments/31_solve_cost/e2e_{chain,chain-sparsest}_p*s*.json`.

2. What ran

Stage A. `examples/e31_stage_a.rs`, one thread: the first 1,000 targets of E25 arm A's stream on each of E25's ten GLS curves ($p = 101$–397, two seeds), the three solvers interleaved per call from the same per-target RNG seed. Record: `experiments/31_solve_cost/stage_a.json`. **End to end.** `experiments/31_solve_cost/run_e2e.sh` (E25's `run_all.sh` plus `--solver`, `RAYON_NUM_THREADS=4`), both solvers, both arms (A: the V_4 fold; B: the $\langle V_4, \psi \rangle$ fold), all ten curves, 16 rho walks per curve. Binary `a8f52422d5017b2e4b219fa2d3324be53b8b6a33` for both. Tables: `experiments/31_solve_cost/tables.md` (`examples/e31_tables.rs`). Host: one shared 4-vCPU cloud container, isolation L0. Every cost is an operation count ($\mathbb{F}_p$

multiplications; a mod-n multiplication at 16); wall times are a practicality note only. Derived values (ratios, the band) are computed from the recorded fields exactly as `e31_tables.rs` does: $S(w) = \text{total_muls} + w \cdot G \cdot \text{macaulay_entries}$, $\rho(w) = \text{rho_matched_muls} + w \cdot G \cdot \text{rho_matched_steps}$, $G = 97$.

3. Stage A: one call, three solvers

p	seed	C med. existing	C med. A	C med. B	exist. / A	exist. / B	NF exist. / A	charpoly exist. / A	eigvec exist. / A	echelon A / B	fallbacks A, B
101	1	1,617,013	600,981	304,257	2.69	5.31	9.2	2.85	16.4	2.52	12, 12
101	2	1,616,974	600,890	304,103	2.69	5.32	9.7	2.87	15.7	2.54	11, 11
149	1	1,632,660	612,064	313,237	2.67	5.21	10.6	2.87	18.7	2.58	7, 7
149	2	1,633,472	612,068	313,320	2.67	5.21	9.5	2.82	15.4	2.54	10, 10
197	1	1,645,617	620,058	320,425	2.65	5.14	10.9	2.88	20.4	2.60	5, 5
197	2	1,646,416	620,152	320,741	2.65	5.13	10.0	2.84	15.6	2.56	8, 8
269	1	1,650,218	623,969	323,602	2.64	5.10	10.9	2.84	20.5	2.60	5, 5
269	2	1,651,643	624,544	324,142	2.64	5.10	11.2	2.85	25.9	2.61	4, 4
397	1	1,663,086	633,960	333,134	2.62	4.99	11.6	2.86	26.5	2.62	3, 3
397	2	1,661,986	633,419	332,624	2.62	5.00	12.4	2.89	30.0	2.65	1, 1

C = Weil restriction + solve per call; phase ratios are of per-call means over all calls, fallbacks charged. M_{P_2} and the characteristic polynomial agreed on 9,996 of 9,996 calls the existing solver does not refuse. Fallbacks (charged reruns of the existing solver): 0.1–1.2 % of calls per curve. Wall median existing / A: 1.56–1.67.

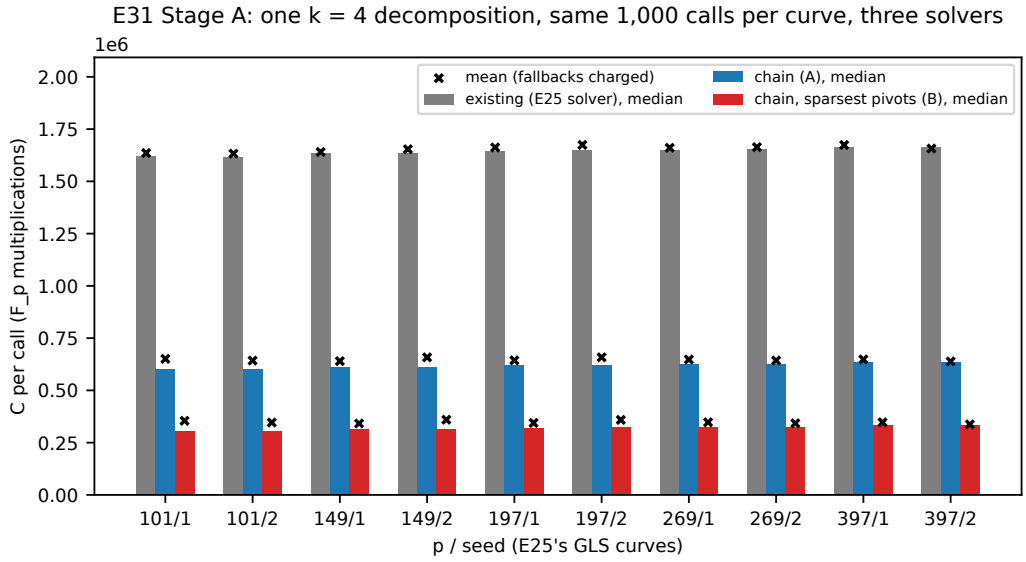


Figure 2: Median (bar) and mean (×) C per call, 1,000 calls per curve, three solvers on the same calls; `experiments/31_solve_cost/stage_a.json`.

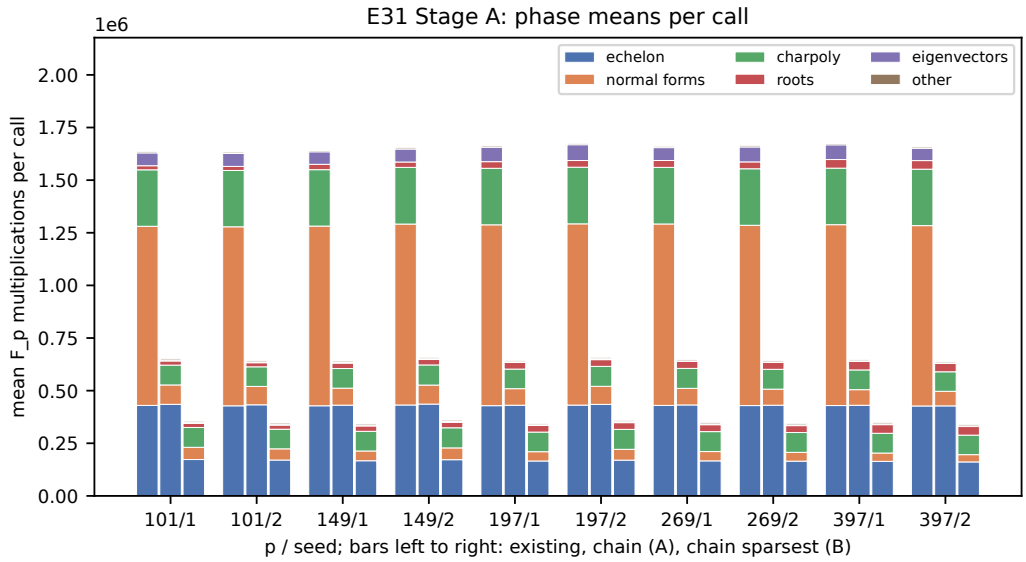


Figure 3: Phase means per call (all calls, fallbacks charged): existing, A, B left to right per curve; `stage_a.json`.

4. End to end: E25's curves rerun

p	seed	arm	S E25	S A	S B	E25 / A	E25 / B	S/ρ E25	S/ρ A	S/ρ B
101	1	A	$6.76 \cdot 10^{10}$	$2.71 \cdot 10^{10}$	$1.49 \cdot 10^{10}$	2.50	4.53	$1.31 \cdot 10^{5}$	$5.27 \cdot 10^{4}$	$2.90 \cdot 10^{4}$
101	1	B	$3.48 \cdot 10^{10}$	$1.39 \cdot 10^{10}$	$7.63 \cdot 10^{9}$	2.51	4.57	$6.77 \cdot 10^{4}$	$2.70 \cdot 10^{4}$	$1.48 \cdot 10^{4}$
101	2	A	$2.47 \cdot 10^{10}$	$9.87 \cdot 10^{9}$	$5.42 \cdot 10^{9}$	2.50	4.56	$6.19 \cdot 10^{4}$	$2.47 \cdot 10^{4}$	$1.36 \cdot 10^{4}$
101	2	B	$1.17 \cdot 10^{10}$	$4.68 \cdot 10^{9}$	$2.58 \cdot 10^{9}$	2.50	4.54	$2.93 \cdot 10^{4}$	$1.17 \cdot 10^{4}$	$6.46 \cdot 10^{3}$
149	1	A	$5.83 \cdot 10^{11}$	$2.30 \cdot 10^{11}$	$1.25 \cdot 10^{11}$	2.53	4.66	$7.07 \cdot 10^{5}$	$2.79 \cdot 10^{5}$	$1.52 \cdot 10^{5}$
149	1	B	$2.19 \cdot 10^{11}$	$8.64 \cdot 10^{10}$	$4.70 \cdot 10^{10}$	2.53	4.65	$2.65 \cdot 10^{5}$	$1.05 \cdot 10^{5}$	$5.71 \cdot 10^{4}$
149	2	A	$1.01 \cdot 10^{11}$	$3.98 \cdot 10^{10}$	$2.17 \cdot 10^{10}$	2.53	4.64	$1.24 \cdot 10^{5}$	$4.92 \cdot 10^{4}$	$2.68 \cdot 10^{4}$
149	2	B	$5.67 \cdot 10^{10}$	$2.25 \cdot 10^{10}$	$1.23 \cdot 10^{10}$	2.52	4.62	$7.01 \cdot 10^{4}$	$2.78 \cdot 10^{4}$	$1.52 \cdot 10^{4}$
197	1	A	$8.53 \cdot 10^{10}$	$3.35 \cdot 10^{10}$	$1.83 \cdot 10^{10}$	2.54	4.67	$8.24 \cdot 10^{4}$	$3.24 \cdot 10^{4}$	$1.76 \cdot 10^{4}$
197	1	B	$6.32 \cdot 10^{10}$	$2.49 \cdot 10^{10}$	$1.36 \cdot 10^{10}$	2.54	4.66	$6.11 \cdot 10^{4}$	$2.40 \cdot 10^{4}$	$1.31 \cdot 10^{4}$
197	2	A	$4.39 \cdot 10^{10}$	$1.72 \cdot 10^{10}$	$9.37 \cdot 10^{9}$	2.55	4.68	$4.44 \cdot 10^{4}$	$1.74 \cdot 10^{4}$	$9.48 \cdot 10^{3}$
197	2	B	$2.59 \cdot 10^{10}$	$1.02 \cdot 10^{10}$	$5.53 \cdot 10^{9}$	2.55	4.68	$2.62 \cdot 10^{4}$	$1.03 \cdot 10^{4}$	$5.60 \cdot 10^{3}$
269	1	A	$1.37 \cdot 10^{11}$	$5.36 \cdot 10^{10}$	$2.90 \cdot 10^{10}$	2.56	4.73	$8.41 \cdot 10^{4}$	$3.29 \cdot 10^{4}$	$1.78 \cdot 10^{4}$
269	1	B	$7.88 \cdot 10^{10}$	$3.09 \cdot 10^{10}$	$1.67 \cdot 10^{10}$	2.55	4.72	$4.84 \cdot 10^{4}$	$1.90 \cdot 10^{4}$	$1.02 \cdot 10^{4}$
269	2	A	$5.10 \cdot 10^{10}$	$1.99 \cdot 10^{10}$	$1.07 \cdot 10^{10}$	2.56	4.76	$3.16 \cdot 10^{4}$	$1.23 \cdot 10^{4}$	$6.65 \cdot 10^{3}$
269	2	B	$2.52 \cdot 10^{10}$	$9.83 \cdot 10^{9}$	$5.30 \cdot 10^{9}$	2.56	4.75	$1.56 \cdot 10^{4}$	$6.09 \cdot 10^{3}$	$3.29 \cdot 10^{3}$
397	1	A	$7.19 \cdot 10^{10}$	$2.81 \cdot 10^{10}$	$1.52 \cdot 10^{10}$	2.56	4.72	$1.95 \cdot 10^{4}$	$7.62 \cdot 10^{3}$	$4.13 \cdot 10^{3}$
397	1	B	$4.17 \cdot 10^{10}$	$1.63 \cdot 10^{10}$	$8.82 \cdot 10^{9}$	2.56	4.73	$1.13 \cdot 10^{4}$	$4.42 \cdot 10^{3}$	$2.39 \cdot 10^{3}$
397	2	A	$1.31 \cdot 10^{11}$	$5.15 \cdot 10^{10}$	$2.79 \cdot 10^{10}$	2.55	4.70	$3.97 \cdot 10^{4}$	$1.56 \cdot 10^{4}$	$8.45 \cdot 10^{3}$
397	2	B	$5.65 \cdot 10^{10}$	$2.22 \cdot 10^{10}$	$1.20 \cdot 10^{10}$	2.55	4.70	$1.71 \cdot 10^{4}$	$6.71 \cdot 10^{3}$	$3.64 \cdot 10^{3}$

At 0 GAE. Replay: every E31 record matches its E25 record on the fields `n`, `residuals`, `unsolved`, `reruns`, `decompositions`, `lifted`, `off_base`, `check_failures`, `rational_solutions`, `base_muls`, `generation_muls`, `weil_muls`, `lift_muls`, `check_muls`, `la_fp_muls`, `rho_matched_muls`, `rho_matched_steps`, `folded`, `rho` (all identical); fallbacks equal E25's `reruns + unsolved` in every record (yes); every log and walk correct (yes). Weil + solves are 98.4–98.9 % of S (A) and 97.0–97.9 % (B).

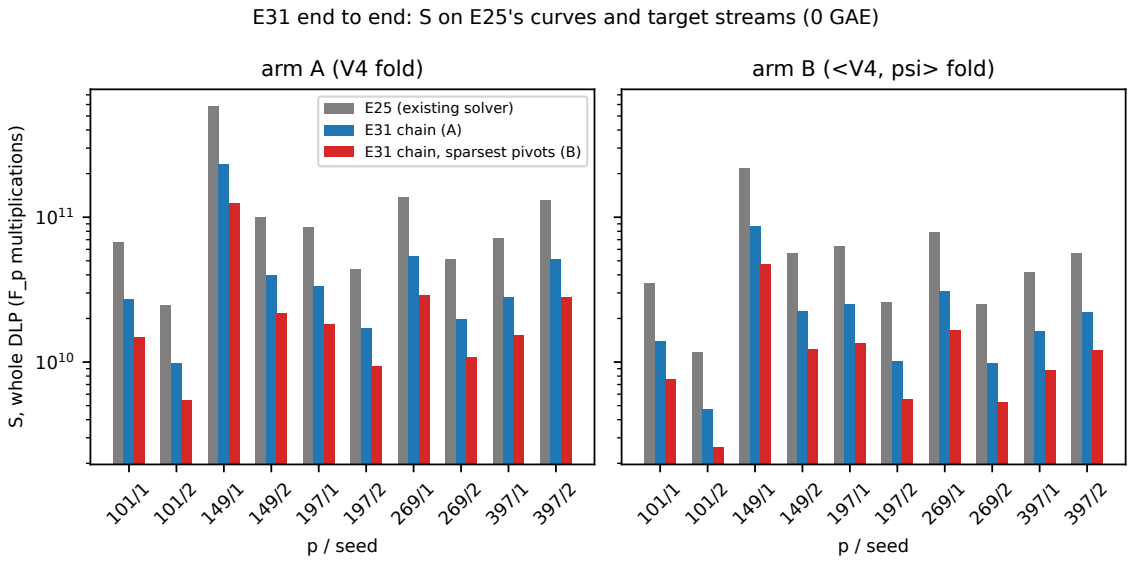


Figure 4: S per curve, E25 against E31 A and B, both arms, 0 GAE.

4.1. The §5a band (ranges over all 20 curve–arm rows)

GAE per native op	S_{E25}/S_A	S_{E25}/S_B	S/ρ E25	S/ρ A	S/ρ B
0	2.50–2.56	4.53–4.76	$1.13 \cdot 10^4$–$7.07 \cdot 10^5$	$4.42 \cdot 10^3$–$2.79 \cdot 10^5$	$2.39 \cdot 10^3$–$1.52 \cdot 10^5$
0.1	1.39–1.43	1.59–1.64	$2.15 \cdot 10^4$–$1.38 \cdot 10^6$	$1.50 \cdot 10^4$–$9.77 \cdot 10^5$	$1.31 \cdot 10^4$–$8.55 \cdot 10^5$
1	1.034–1.053	1.051–1.071	$7.72 \cdot 10^4$–$5.45 \cdot 10^6$	$7.33 \cdot 10^4$–$5.22 \cdot 10^6$	$7.21 \cdot 10^4$–$5.14 \cdot 10^6$

The order E31 < E25 holds at every weight on every row, and so does rho < E31: both rankings are admitted by §5a. The size of the E31 gain is not band-stable (it shrinks to a few percent at 1 GAE).

5. Predictions and verdicts

Registered bands are quoted from §9.22; measured values from the records.

id	registered	measured → verdict
G2.85	identical outputs on every call	Stage A 10,000 / 10,000 (A), 10,000 / 10,000 (B); end to end identical → holds
G2.86	NF existing / A in $[12, 40]$	9.2–12.4 (fallbacks charged) → falsified low
G2.87	charpoly existing / A in $[1.8, 3.2]$	2.82–2.89 → holds
G2.88	eigenvectors existing / A in $[15, 80]$	15.4–30.0 → holds
G2.89	C median existing / A in $[2.2, 3.0]$; kill < 1.5	2.62–2.69 → holds
G2.90	fallbacks $\leq 1\%$ of calls	Stage A 0.1–1.2 %; end to end 0.27–1.20 % → falsified at $p = 101$
G2.91	wall existing / A in $[1.2, 2.0]$	1.56–1.67 → holds
G2.92	echelon A / B in $[1.0, 1.6]$ (exploratory)	2.52–2.65 → falsified high; B carried end to end
G2.93	S_{E25}/S_A in $[2.1, 2.9]$, counts reproduced	2.50–2.56 → holds
G2.94	$[1.3, 1.6]$ at 0.1, $[1.03, 1.10]$ at 1 GAE; A below E25	1.39–1.43 and 1.034–1.053 → holds

6. Cross-check of §9.23 against the records

Every value in §9.23's Stage A and end-to-end tables, the §5a ranges, the agreement counts and the §9.22 profile table agree with the JSON records at the precision printed. Secondary statements, re-read from the records (`summary.json`, `cross_checks`):

- **Fallback rate per end-to-end run** (§9.23 G2.90 row and "next levers": "`0.2–1.1 %`"): the 40 records give 0.27–1.20 % (fallbacks / calls), i.e. about 0.3–1.2 %. The upper end is 1.2 %, not 1.1 % (e.g. 85 / 7,106 at $p = 101$ seed 2 arm B, 488 / 41,043 at $p = 101$ seed 1 arm A). Verdict unchanged.
- **Cost of a fallback call to the existing solver** ("`4.5–4.8·10⁶`"): the Stage A calls identified as fallbacks (rule: chain normal_forms > 300000 finds 62 of 66 recorded) cost $4.44 \cdot 10^6$–$4.99 \cdot 10^6$. The chain attempt adds 13.1 % (§9.23: ≈ 13 %).
- **A's echelon share** ("`≈ 70 %`"): 66.1–67.4 % of the mean C; charpoly 14.4–14.7 % (§9.23: ≈ 15 %).
- **Fallback weight in B at $p = 101$** ("up to `≈ 16 %` of B's mean C"): the identified fallback calls carry at most 15.0 % of B's Stage A multiplications (approximate: Stage A records no per-call fallback flag).
- Clean-call normal forms: A $6.56 \cdot 10^4$–$6.68 \cdot 10^4$, B $3.16 \cdot 10^4$–$3.22 \cdot 10^4$ (§9.23: `65.6–66.8·10³`, `31.6–32.2·10³`): agree.
- "every row is still `2.39·10³–1.52·10⁵×`" is B's range; A's rows reach $2.79 \cdot 10^5\times$.

7. Scope and limits

E25's GLS twists over $\mathbb{F}_{p^4}$, $p \leq 397$, $n \leq 2^{30.5}$, the degree-11 solve; E24's curves use the same solver but were not rerun end to end. Operation counts only, on an L0 shared host. The §11.16 handover was not recomputed. Validation gate: the replay certificate, validator report and `VALIDATION.json` for these rows are prepared separately in `benchmarks/20261024-e31-chain-solve/`; until they land, the rows are unvalidated and do not enter the leaderboard.

8. Records

Raw: `experiments/31_solve_cost/` (`stage_a.json`, `e2e_chain_p*s*.json`, `e2e_chain-sparsest_p*s*.json`, logs, `profile_p269_p397.json`), `experiments/30_gls_v4/p*s*.json` (E25). Code: `klein_v4::solve_weighted_chain` beside `solve_weighted_traced`; runners `examples/e31_stage_a.rs`, `examples/quartic_pipeline_gls.rs --solver`, `examples/e31_tables.rs`. Figures and this report's numbers: `research/e31_chain_solve_20261024/figures/plot.py` → `summary.json`.